

The slide features a vertical stack of seven horizontal bars on the left side, alternating between white and blue. The top bar is white, followed by a blue one, and so on, creating a descending staircase pattern. The main content area has a dark grey background with white text. At the top left, there's a logo for NOVA IMS Information Management School. The title 'Data Mining' is prominently displayed in the upper right. Below it, 'Profiling' is written in a larger font, followed by 'NOVA-IMS'. The presenter's name, 'Fernando Lucas Bação', is listed below that, along with his email address 'fbacao@isegi.unl.pt' and website 'http://www.isegi.unl.pt/fbacao'. At the bottom, the affiliation 'Instituto Superior de Estatística e Gestão de Informação Universidade Nova de Lisboa' is provided.

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Clustering

- **Profiling the solution**
 - Analyst evaluation is a critical step in the clustering process, where the clustering results are analyzed and evaluated by a domain expert or data analyst.
 - Here are some reasons why analyst evaluation is important:
 - Ensuring relevance: should help ensure that the clusters make sense in the context of the business problem or research question being addressed. Clustering algorithms produce clusters that are technically accurate but may not be relevant to the problem at hand. Analyst evaluation can help identify and discard irrelevant clusters.

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Clustering

- **Profiling the solution**
 - Analyst evaluation is a critical step in the clustering process, where the clustering results are analyzed and evaluated by a domain expert or data analyst.
 - Here are some reasons why analyst evaluation is important:
 - Identifying anomalies: Analyst evaluation can help identify and analyze data points that do not fit within any of the identified clusters. These outliers or anomalies may be of particular interest to the analyst or may indicate issues with the clustering algorithm or the data.
 - Improving interpretability: Clustering algorithms may produce clusters that are difficult to interpret or understand. Analyst evaluation can help identify and explain the underlying patterns or characteristics of the clusters and provide insights that can be easily communicated to stakeholders.

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Clustering

- **Profiling the solution**
 - Analyst evaluation is a critical step in the clustering process, where the clustering results are analyzed and evaluated by a domain expert or data analyst.
 - Here are some reasons why analyst evaluation is important:
 - Refining the clustering solution: Analyst evaluation can help refine the clustering solution by suggesting modifications to the clustering algorithm or the input data. For example, the analyst may suggest using a different feature set, or may identify missing or noisy data that needs to be cleaned or imputed.
 - Validating the results: Analyst evaluation can provide a form of validation for the clustering results. By comparing the results to prior knowledge or expert opinion, the analyst can help ensure that the clusters are meaningful and useful.

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Clustering

- **Profiling the solution**
 - Analyst evaluation is a critical step in the clustering process, where the clustering results are analyzed and evaluated by a domain expert or data analyst.
 - Overall, analyst evaluation plays an important role in the clustering process by ensuring that the clustering results are relevant, interpretable, and valid.
 - It allows for a more thorough understanding of the data and the patterns it contains, and can lead to more actionable insights and better decision-making.

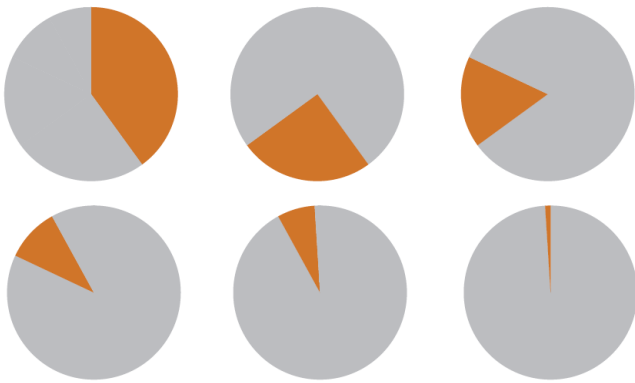
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Clustering

- Profiling (size of the clusters)**
Part-to-Whole Mini-Pie Charts



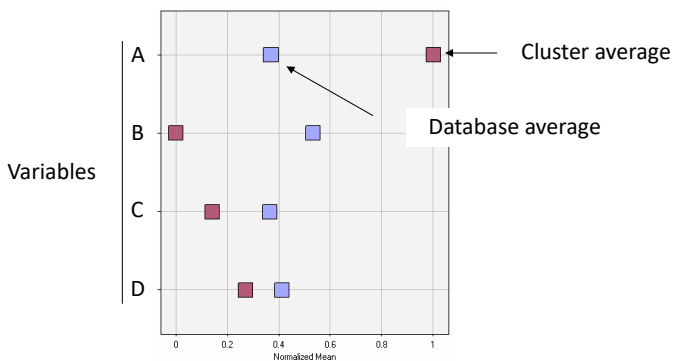
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Clustering

- Profiling (comparing averages)**



Variables

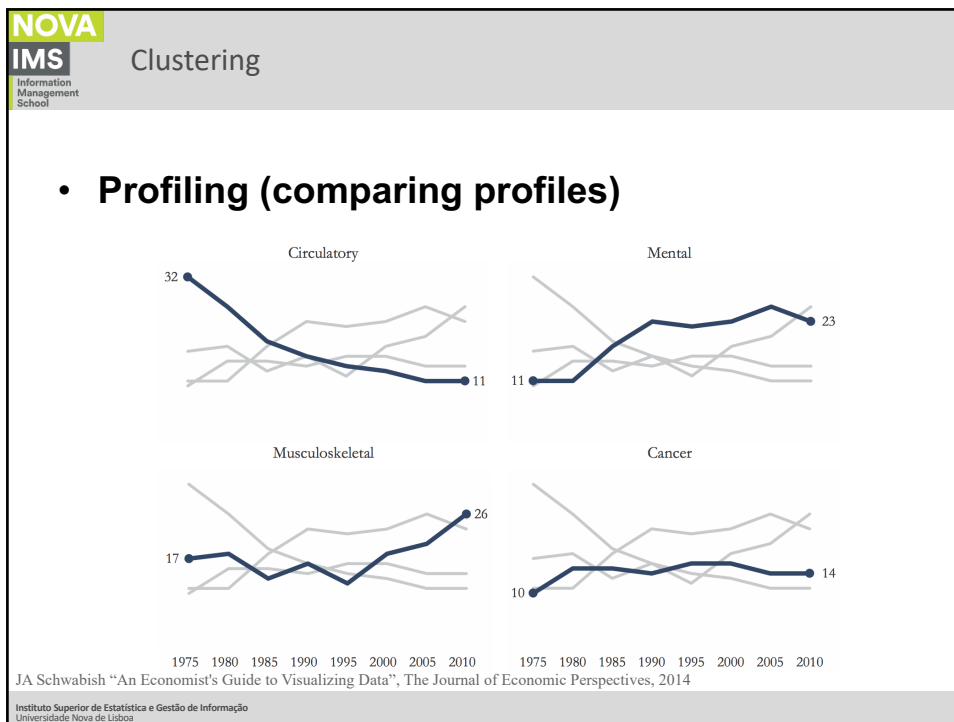
Cluster average

Database average

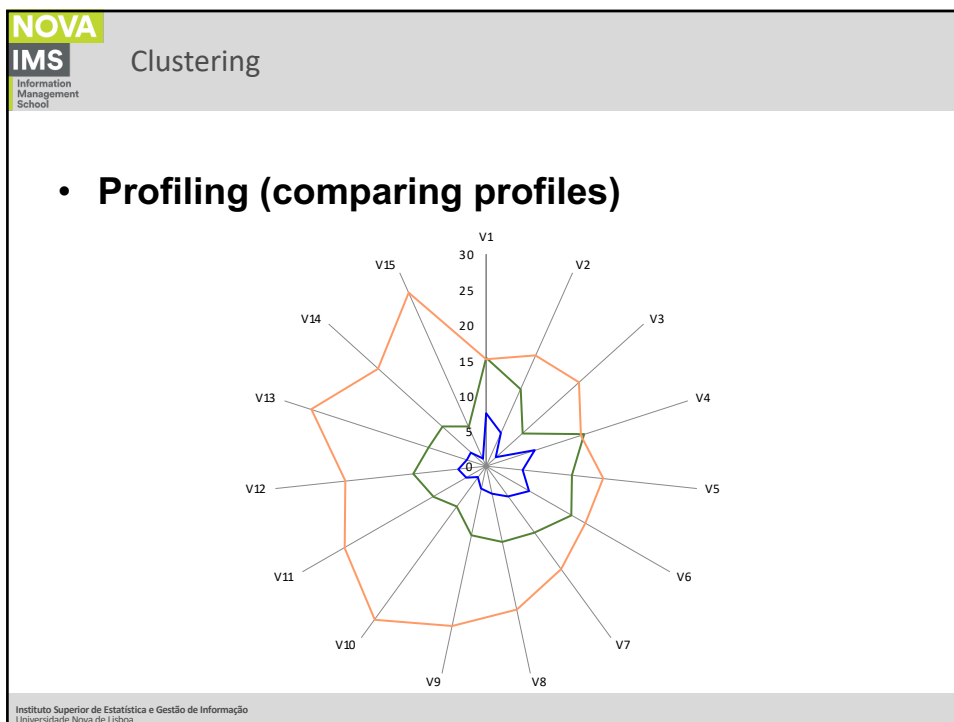
Normalized Mean

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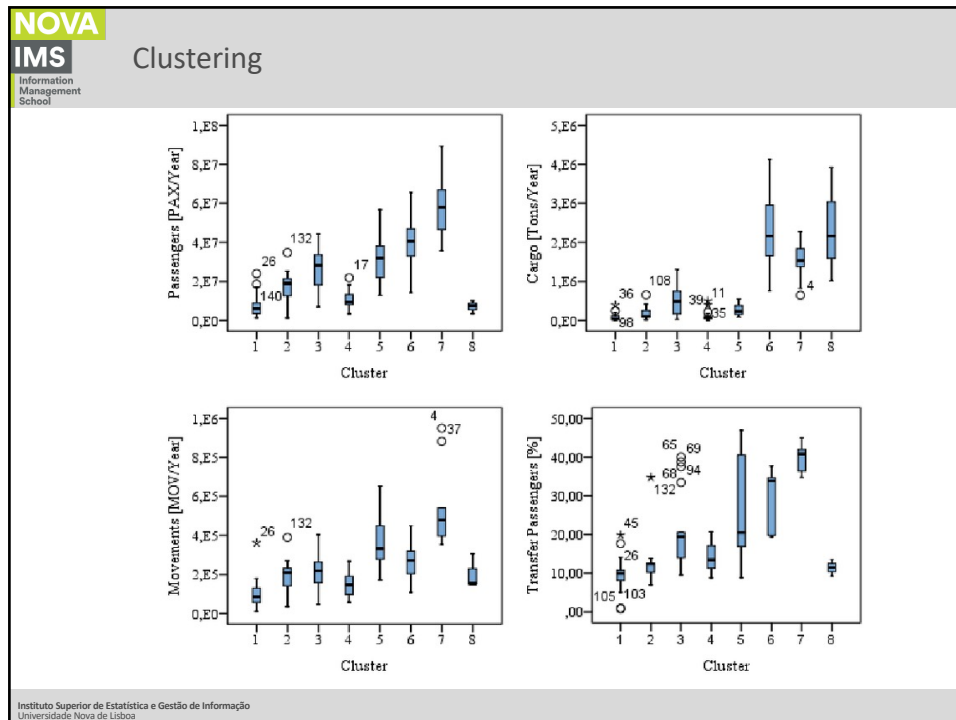
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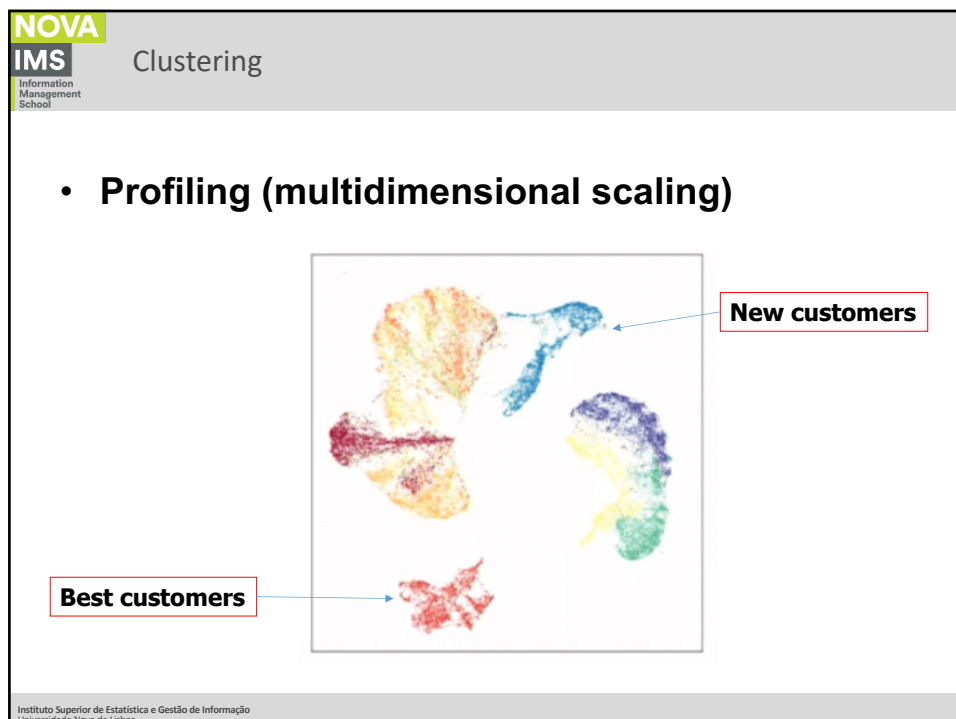
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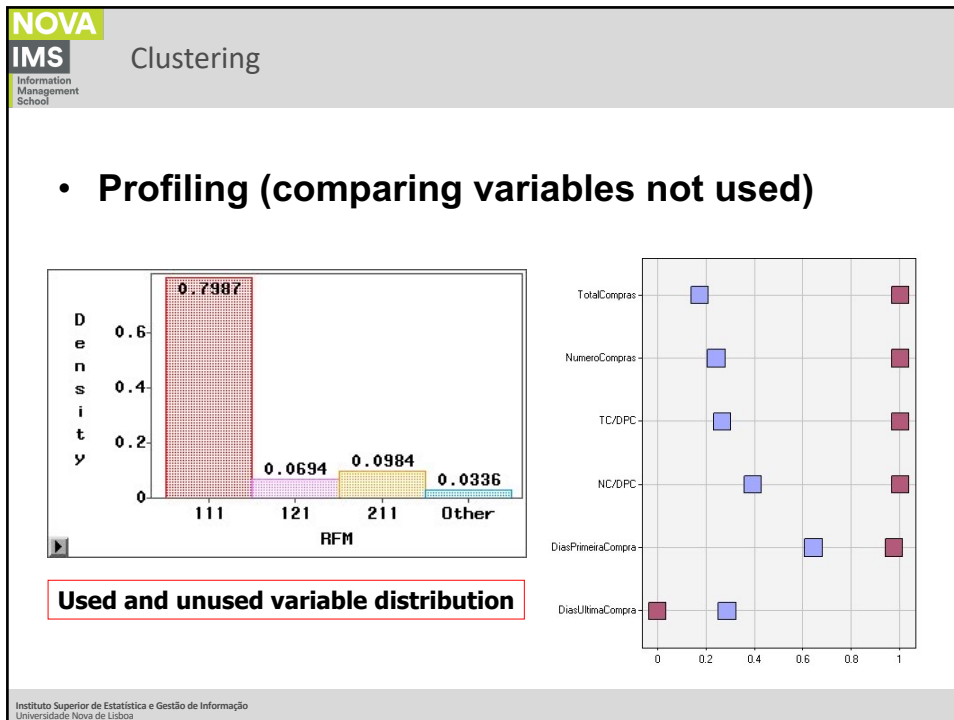
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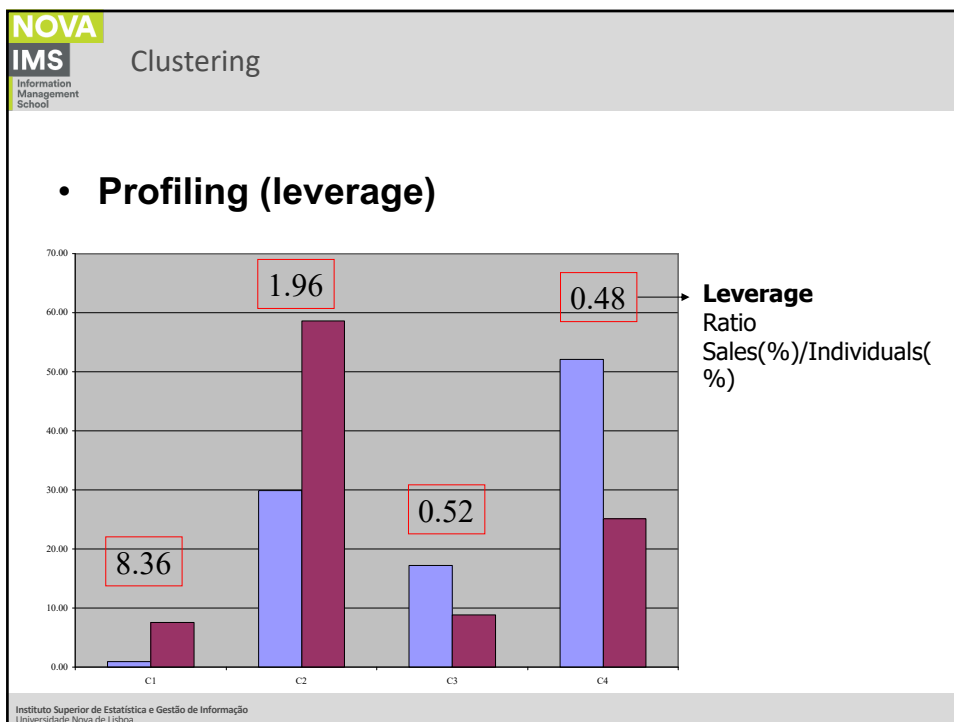
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Clustering

- Exploring two solutions

		Sofist Digital	Variedade Serviço	Preço	Status_PC	ProAtividade	Tempo Internet e RS	Sustentabilidade	Life Status	Propensao_Pfi delizacao	Switching	
k=5	FREQ	F1_SD	F2_VS	F3_P	F4_SPC	F5_PA	F6_IRS	F7_CC	F8_LS	F9_FI	F10_AB	
1	222	-0,40	-0,54	-0,31	-0,29	0,13	0,65	-1,67	-0,20	-0,09	-0,03	Pouco Envolvidos e pouco sensíveis ao preço
2	463	0,70	0,44	0,38	0,13	-0,32	-0,37	-0,19	0,31	-0,58	0,03	Social Customers
3	469	0,58	-0,16	0,15	0,48	0,23	0,00	0,09	-0,22	0,98	-0,04	Sofisticado com Propensao Fidelizacao
4	428	-1,47	0,27	0,25	-0,13	0,01	-0,25	0,29	0,15	0,15	0,04	Info Excluidos
5	420	0,29	-0,30	-0,68	-0,39	0,02	0,31	0,69	-0,16	-0,56	-0,02	Sustentaveis
k=6	FREQ	F1_SD	F2_VS	F3_P	F4_SPC	F5_PA	F6_IRS	F7_CC	F8_LS	F9_FI	F10_AB	
1	241	0,12	-0,30	-0,09	-0,20	-0,04	1,34	-1,29	-0,03	-0,13	-0,24	Pouco Envolvidos
2	134	-0,16	0,11	0,02	-0,15	-0,10	0,13	0,10	0,00	0,07	2,84	Switchers
3	285	0,43	0,48	-1,28	-0,16	0,52	-0,28	0,05	0,04	0,23	-0,07	Exigentes
4	426	-1,42	0,22	0,30	-0,20	-0,03	-0,27	0,10	0,01	0,21	-0,27	Info Excluidos
5	414	0,39	-0,77	0,20	0,92	0,19	-0,23	0,15	0,21	0,15	-0,16	Status/Conservadores
6	502	0,63	0,30	0,34	-0,37	-0,38	-0,10	0,35	-0,19	-0,39	-0,24	Social Customers

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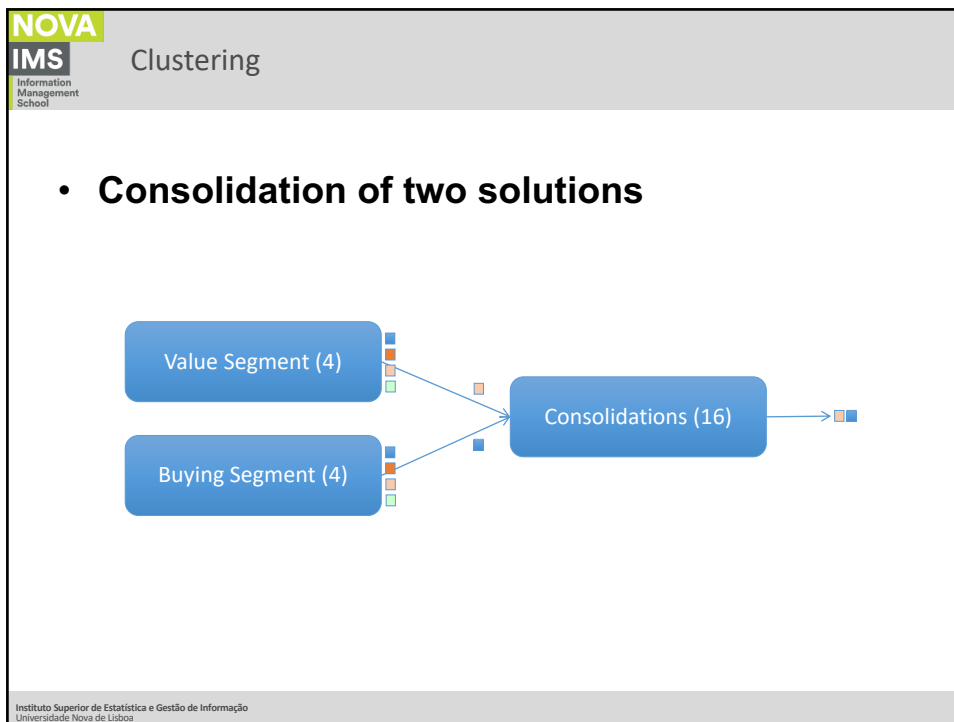
- Exploring two solutions with different k

k=7 VS k=8	Pouco envolvidos	Switchers	Informados / Social Customers	Fidelizáveis Sofisticados	Info Excluidos	Status	Proativos	Desprendido	Grand Total
Pouco envolvidos	71		23	13	45	1	1	49	203
Switchers	2	106	4	4	2	1		3	122
Informados / Social Customers	4		219	68	2	20	1	64	378
Fidelizáveis Sofisticados	72		105	191	4	1	2	69	444
Status/Conservadores	7		5	39	13	157		42	263
Proativos	5		15	20	3		158	19	215
Info Excluidos		12	4	309	1		1	57	377
Grand Total	161	106	383	339	378	181	158	296	2002

k=6 VS k=8	Pouco envolvidos	Switchers	Informados / Social Customers	Fidelizáveis Sofisticados	Info Excluidos	Status	Proativos	Desprendido	Grand Total
Pouco Envolvidos	146		16	24	8	17	7	23	241
Switchers	2	105	6	11	1	2	4	3	134
Exigentes	1	1	26	56	13	27	60	101	285
Info Excluidos			13	22	350	17	11	26	426
Status/Conservadores	3		13	177	5	98	58	60	414
Social Customers	9		322	49	1	20	18	83	502
Grand Total	161	106	383	339	378	181	158	296	2002

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